Pure Mathematics with Mechanics GCE A Level 2023 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0765 Pure Maths With Mechanics 1, June 2023, Advanced Level, one and a half hours). Answer ALL questions; each question has FOUR suggested answers A, B, C and D and only one answer must be marked (if more than one answer is marked the score for that question is zero). Formulae booklets and calculators are allowed. Section A: Pure Mathematics; Section B: Mechanics. Mobile phones are NOT allowed in the examination room.

SECTION A: PURE MATHEMATICS

If $\frac{x}{(x+3)(x+4)} \equiv \frac{P}{x+3} + \frac{Q}{x+4}$, then

a) $P = -3,\ Q = 4$
b) $P = 4,\ Q = -3$
c) $P = 3,\ Q = -4$
d) $P = 3,\ Q = 4$

2. The quadratic equation $\lambda x^2 - 4x + \lambda = 0$ has real and distinct roots only when

a) $16 - 4\lambda^2 < 0$
b) $16 - 4\lambda^2 > 0$
c) $16 - 4\lambda^2 \leq 0$
d) $16 - 4\lambda^2 \geq 0$

3. An equivalence relation, $\mathcal{R}$, is that which is

a) reflexive, symmetric and anti-symmetric
b) transitive, reflexive and symmetric
c) reflexive, injective and symmetric
d) injective, surjective and bijective

4. When the polynomial $P(x) = ax^3 - x^2 - x + 6$ is divided by $(x+1)$, the remainder is 4. The value of the constant a is

a) -2
b) 2
c) -4
d) 4

5. The value of $\frac{1}{3}\log_2 \sqrt{8}$ is

a) $\frac{1}{2}$
b) $\frac{1}{6}$
c) $\frac{3}{5}$
d) $\frac{2}{3}$

6. The sum of the first n terms of a series is given by $S_n = n^2 + 5$. The 5th term of this series is

a) 30
b) 18
c) 9
d) 51

7. $\sum_{r=1}^{4} \left(4^{\frac{1}{2}}\right)^r =$

a) 16
b) 30
c) 32
d) 64

8. Two real valued functions f and g are defined by $f(x) = \frac{2x-3}{x+1},\ x \neq -1$ and $g(x) = \frac{1}{x},\ x \neq 0$. $(g \circ f)(0) =$

a) ∞
b) 0
c) $-\frac{1}{3}$
d) $\frac{1}{3}$

9. The general solution of the differential equation $2y\frac{dy}{dx} = 1 + x$ is

a) $2y = x + \frac{x^2}{2} + k$
b) $2y = x - \frac{x^2}{2} + k$
c) $y^2 = x - \frac{x^2}{2} + k$
d) $y^2 = x + \frac{x^2}{2} + k$

10. The modulus of the complex number $\frac{1+2i}{3-4i}$ is

a) $\frac{\sqrt{4}}{5}$
b) $\frac{\sqrt{5}}{3}$

c) $\frac{\sqrt{5}}{4}$
d) $\frac{\sqrt{5}}{5}$

11. If $(1+2x)^{-1} \equiv 1 - 2x + bx^2$, then the value of b is

a) -2
b) 2
c) -4
d) 4

12. The general solution of the equation $\sin\theta - \cos\theta = 0$ is

a) $n\pi + \frac{\pi}{4}$
b) $n\pi - \frac{\pi}{4}$
c) $2n\pi \pm \frac{\pi}{4}$
d) $2n\pi + \frac{\pi}{4}$

13. The sine of the acute angle between the plane Π and the line L, where $\Pi : x + y + z = 2$ and $L : \frac{x-3}{2} = \frac{y+5}{2} = \frac{z-2}{1}$ is

a) $\frac{-2}{3\sqrt{8}}$
b) $\frac{2}{\sqrt{38}}$
c) $\frac{5}{3\sqrt{3}}$
d) $\frac{5}{\sqrt{3}}$

14. If n is a positive integer, then $\lim_{n\to\infty}\left(\frac{n+3}{2n+1}\right) =$

a) $\frac{1}{2}$
b) 2
c) -3
d) 3

15. $\frac{\sin 5x - \sin x}{\sin 4x - \sin 2x} \equiv$

a) $2\cos 2x$
b) $2\sin x$
c) $2\cos x$
d) $2\sec x$

16. The cofactor of the element 6 in the matrix $\mathbf{M} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ is

a) $\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$
b) $-\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$
c) $\begin{vmatrix} 8 & 7 \\ 2 & 1 \end{vmatrix}$
d) $-\begin{vmatrix} 7 & 8 \\ 1 & 2 \end{vmatrix}$

17. Two statements p and q are given by p: James is dancing; q: James is awake. Then $\sim (p \wedge q)$ means

a) James is not dancing and James is not awake
b) James is not dancing or James is not awake
c) James is not dancing and James is awake
d) James is not dancing or James is awake

18. The diameter of a circle is the line segment AB. If the coordinates of A and B are $(1, 8)$ and $(3, 14)$ respectively, then the equation of the circle is

a) $(x-14)(x-8)+(y-3)(y-1)=0$
b) $(x-3)(x-14)+(y-1)(y-8)=0$
c) $(x-1)(x-14)+(y-3)(y-8)=0$
d) $(x-1)(x-3)+(y-8)(y-14)=0$

19. If $y^2 - y + 2x^2 = 5$, then $\dfrac{dy}{dx} =$

a) $\dfrac{4x}{1-2y}$
b) $\dfrac{4x}{2y-1}$
c) $\dfrac{4x}{1-y}$
d) $\dfrac{4x}{y-1}$

20. The image of the point $(1, 0, 3)$ under the transformation whose matrix is $\mathbf{M}$ where $\mathbf{M} = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 0 & 3 & 1 \end{pmatrix}$ is the point

a) $(1, 3, 3)$
b) $(0, 3, 3)$
c) $(1, 0, 0)$
d) $(3, 0, 3)$

21. If the vectors $\mathbf{r} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{s} = 5\mathbf{i} - a\mathbf{j} + 2\mathbf{k}$ are perpendicular then the value of a is

a) -12
b) -8
c) 8
d) 12

22. Given that $y = x^2 - \dfrac{1}{x}$, $\dfrac{dy}{dx} =$

a) $2x - \dfrac{1}{x^2}$
b) $2x + 1$
c) $2x + \dfrac{1}{x^2}$
d) $2x - 1$

23. $\displaystyle\int \frac{x}{1+3x^2}\,dx =$

a) $6\ln(1+3x^2) + k$
b) $\dfrac{1}{6}\ln(1+3x^2) + k$
c) $\dfrac{1}{3}\ln(1+3x^2) + k$
d) $3\ln(1+3x^2) + k$

24. A real valued function f is defined by $f(x) = \dfrac{1}{x-2},\ x \neq 2$. The range of f is

a) $\{x \in \mathbb{R} : x \neq 2\}$
b) $\{x \in \mathbb{R} : x \neq -2\}$
c) $\{x \in \mathbb{R} : x \neq 0\}$
d) $\{x \in \mathbb{R} : x \neq 1\}$

25. The range of values of x for which $\dfrac{x+5}{x-3} < 0$ is

a) $x < -5$ or $x > 3$
b) $x < 3$ or $x > 5$
c) -5 d) 3

26. The complex number i^{102} is equal to

a) 1
b) -1
c) i
d) $-i$

27. Given that -1.5 is the first approximation to the root of the equation $f(x) = 0$, where $f : \mathbb{R} \to \mathbb{R}$ is a function, a second approximation of the root of the equation is

a) $-1.5 - \dfrac{f'(-1.5)}{f(-1.5)}$
b) $-1.5 + \dfrac{f(-1.5)}{f'(-1.5)}$
c) $-1.5 - \dfrac{f(-1.5)}{f'(-1.5)}$
d) $-1.5 + \dfrac{f'(-1.5)}{f(-1.5)}$

28. If the set $B = \{4, 5, 8, 10\}$, the cardinality of the power set of B is

a) 4
b) 14

c) 16
d) 8

29. $\int_0^{\frac{\pi}{2}} 6\cos 2x\,dx =$

a) 3
b) 0
c) 2
d) 1

30. The parametric equations of a curve are $x = 3t + \sin 3t$ and $y = 3 - \cos 3t$. $\dfrac{dy}{dx} =$

a) $\dfrac{\sin 3t}{3 + \cos 3t}$
b) $\dfrac{\sin 3t}{1 + 3\cos 3t}$
c) $\dfrac{\sin 3t}{1 + \cos 3t}$
d) $\dfrac{\sin 3t}{3 + \cos t}$

31. The equations of the three asymptotes of the curve $y = \dfrac{x-2}{(x+5)(x-1)}$ are

a) $y = 1,\ x = 5,\ x = 2$
b) $y = 0,\ x = -5,\ x = 1$
c) $y = 0,\ x = 5,\ x = 1$
d) $y = 1,\ x = -5,\ x = 1$

32. $\dfrac{\cos 5\theta - i\sin 5\theta}{\cos 3\theta - i\sin 3\theta} =$

a) $\cos 2\theta - i\sin 2\theta$
b) $\cos 2\theta + i\sin 2\theta$
c) $\cos 8\theta - i\sin 8\theta$
d) $\cos 8\theta + i\sin 8\theta$

33. A committee of 6 is to be formed from 10 men and 6 women. In how many ways can this selection be done such that there should be an equal number of men and women?

a) ${}^{10}C_3 + {}^{6}C_3$
b) ${}^{10}C_6 \cdot {}^{6}C_6$
c) ${}^{3}C_{10} + {}^{3}C_6$
d) ${}^{10}C_3 \cdot {}^{6}C_3$

34. The table below gives values of y corresponding to some given values of x. Using the trapezium rule, an approximate value for $\int_1^5 y\,dx$ is

(The table of values announced in the question is not reproduced in the scanned paper: the page shows no table.)

Options as printed: A) 32 B) 36 C) 16 D) 28

35. $\int \sin 8x \cos 2x \, dx =$

a) $-\left(\dfrac{\cos 10x}{10} + \dfrac{\cos 6x}{6}\right) + K$
b) $\left(\dfrac{\cos 10x}{10} + \dfrac{\cos 6x}{6}\right) + K$
c) $-\dfrac{1}{2}\left(\dfrac{\cos 10x}{10} + \dfrac{\cos 6x}{6}\right) + K$
d) $\dfrac{1}{2}\left(\dfrac{\cos 10x}{10} + \dfrac{\cos 6x}{6}\right) + K$

SECTION B: MECHANICS

The position vector of a particle is $\mathbf{r} = \left[(t^3 + 2t^2 + 2)\mathbf{i} + (t^2 - 1)\mathbf{j}\right]$ m at time t s. The acceleration of the particle when $t = 1$ is

a) $(10\mathbf{i} - 2\mathbf{j})\ \mathrm{m\,s^{-2}}$
b) $(2\mathbf{i} + 10\mathbf{j})\ \mathrm{m\,s^{-2}}$
c) $(7\mathbf{i} + 2\mathbf{j})\ \mathrm{m\,s^{-2}}$
d) $(10\mathbf{i} + 2\mathbf{j})\ \mathrm{m\,s^{-2}}$

37. A particle is projected from a point on a horizontal plane with speed $10\ \mathrm{m\,s^{-1}}$ at an angle 30° to the plane. Taking g as $10\ \mathrm{m\,s^{-2}}$, the greatest height H attained by the particle is

a) $H = 5\ \mathrm{m}$
b) $H = 1.25\ \mathrm{m}$
c) $H = \dfrac{5}{2}\sqrt{3}\ \mathrm{m}$
d) $H = 5\sqrt{3}\ \mathrm{m}$

38. The work that must be done to compress a spring of modulus $500\ \mathrm{N}$ from its natural length of $0.1\ \mathrm{m}$ to a shortened length of $0.08\ \mathrm{m}$ is

a) $1\ \mathrm{J}$
b) $0.1\ \mathrm{J}$
c) $2\ \mathrm{J}$
d) $50\ \mathrm{J}$

39. A particle of mass $4\ \mathrm{kg}$ is moving in a horizontal circle of radius $2\ \mathrm{m}$ with angular speed of $3\ \mathrm{rad\,s^{-1}}$. The centripetal force acting on the particle is

a) $18\ \mathrm{N}$
b) $72\ \mathrm{N}$
c) $24\ \mathrm{N}$
d) $3\ \mathrm{N}$

40. Three particles of weight $7\ \mathrm{N}$, $9\ \mathrm{N}$ and $4\ \mathrm{N}$ have position vectors $(4\mathbf{i} + \mathbf{j})\ \mathrm{m}$, $3\mathbf{j}\ \mathrm{m}$ and $(3\mathbf{i} + 4\mathbf{j})\ \mathrm{m}$ respectively. The position vector of their center of mass is

a) $\left(\dfrac{7}{3}\mathbf{i} + \dfrac{8}{3}\mathbf{j}\right)\ \mathrm{m}$
b) $\left(2\mathbf{i} + \dfrac{5}{2}\mathbf{j}\right)\ \mathrm{m}$

c) $(40\mathbf{i} + 50\mathbf{j})$ m
d) $\left(\frac{7}{20}\mathbf{i} + \frac{8}{20}\mathbf{j}\right)$ m

41. Fig. 1 shows a uniform sheet of metal in the form of the letter T with dimensions in meters.

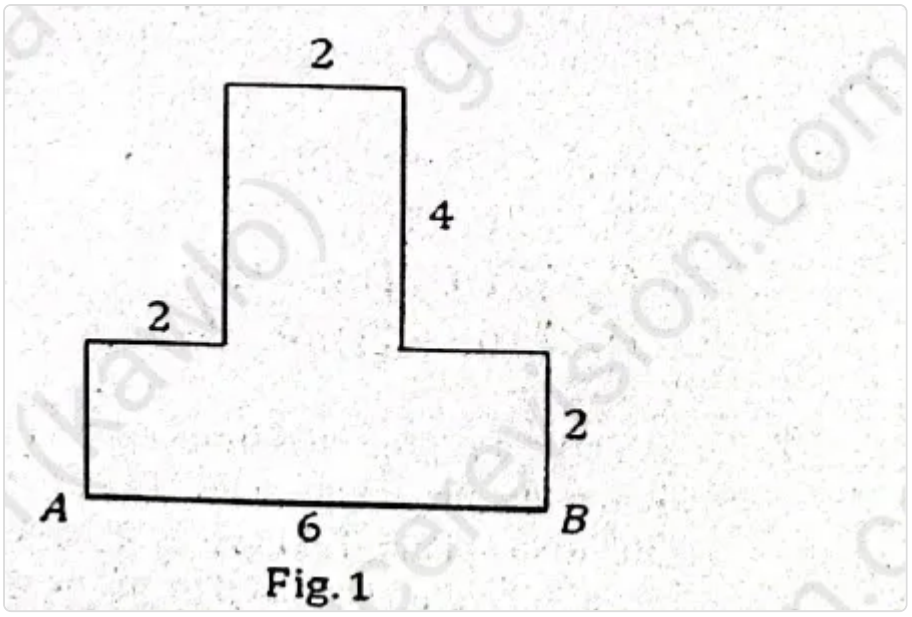


Fig. 1

The distance of the center of mass of the sheet from the edge AB is

a) 2.5 m
b) 1.3 m
c) 2.2 m
d) 3 m

42. A uniform ladder AB of length $2a$ rests with its upper end B against a rough vertical wall and the lower end A against a rough horizontal floor. The forces acting on the ladder AB are shown in fig. 2.

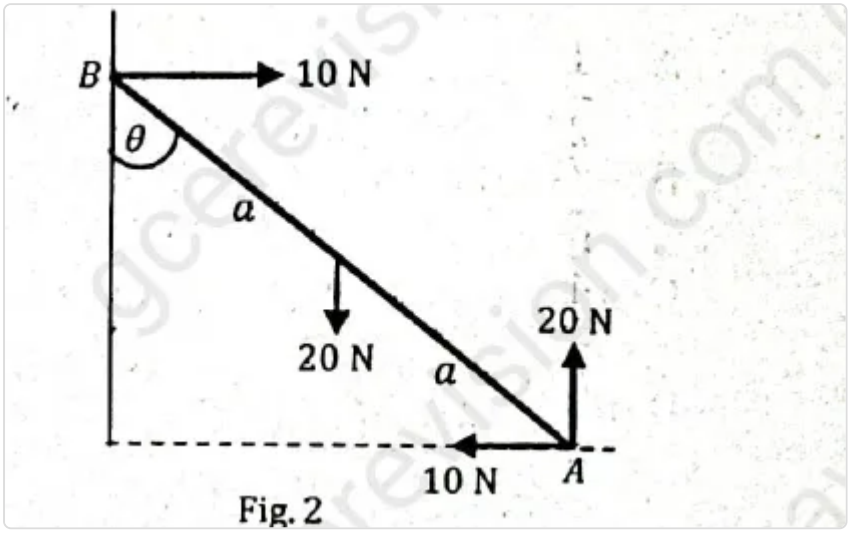


Fig. 2

Given that AB is in limiting equilibrium, the angle θ which AB makes with the wall is

a) 35°
b) 30°
c) 60°
d) 45°

43. X, Y and Z are moving particles. The velocity of X is $(4\mathbf{i}+5\mathbf{j})\ \mathrm{m\,s^{-1}}$. Relative to X the velocity of Y is $(3\mathbf{i}+2\mathbf{j})\ \mathrm{m\,s^{-1}}$ and relative to Y the velocity of Z is $(5\mathbf{i}-3\mathbf{j})\ \mathrm{m\,s^{-1}}$. The true velocity of Z is

a) $(12\mathbf{i}+4\mathbf{j})\ \mathrm{m\,s^{-1}}$
b) $(9\mathbf{i}+2\mathbf{j})\ \mathrm{m\,s^{-1}}$
c) $(7\mathbf{i}+7\mathbf{j})\ \mathrm{m\,s^{-1}}$
d) $(8\mathbf{i}-\mathbf{j})\ \mathrm{m\,s^{-1}}$

44. A particle moves so that after t seconds its position vector $\mathbf{r}=\left[(4t-1)\mathbf{i}+t^2\mathbf{j}+(15t-t^3)\mathbf{k}\right]\ \mathrm{m}$. The speed of the particle after 3 seconds is

a) $(4\mathbf{i}+6\mathbf{j}-12\mathbf{k})\ \mathrm{m\,s^{-1}}$
b) $(3\mathbf{i}+6\mathbf{j}-12\mathbf{k})\ \mathrm{m\,s^{-1}}$
c) $14\ \mathrm{m\,s^{-1}}$
d) $\sqrt{189}\ \mathrm{m\,s^{-1}}$

45. A sphere of mass $2\ \mathrm{kg}$ collides directly with another sphere of mass $8\ \mathrm{kg}$. Both spheres are moving in the same direction with speeds $3\ \mathrm{m\,s^{-1}}$ and $1\ \mathrm{m\,s^{-1}}$ respectively. Given that the $2\ \mathrm{kg}$ mass is brought to rest by the impact, the coefficient of restitution between the spheres is

a) $\dfrac{1}{2}$
b) $\dfrac{7}{8}$
c) $\dfrac{4}{7}$
d) $\dfrac{3}{7}$

46. The work done by a force $\mathbf{F}=(\mathbf{i}+2\mathbf{j}+3\mathbf{k})\ \mathrm{N}$ which moves its point of application from a point P with position vector $\mathbf{r}_P=(3\mathbf{i}+4\mathbf{j}+5\mathbf{k})\ \mathrm{m}$ to another point Q with position vector $\mathbf{r}_Q=(7\mathbf{i}+6\mathbf{j}+5\mathbf{k})\ \mathrm{m}$ is

a) $-60\ \mathrm{J}$
b) $60\ \mathrm{J}$
c) $-8\ \mathrm{J}$
d) $8\ \mathrm{J}$

47. A car of mass $1200\ \mathrm{kg}$ moving with a speed of $12\ \mathrm{m\,s^{-1}}$ has its engine working at $54\ \mathrm{kW}$ up a road inclined at an angle $\sin^{-1}\left(\dfrac{1}{5}\right)$ to the horizontal. The non-gravitational resistance to the motion of the car is $300\ \mathrm{N}$. Taking g as $10\ \mathrm{m\,s^{-2}}$, the acceleration of the car is

a) $2.5\ \mathrm{m\,s^{-2}}$
b) $1.5\ \mathrm{m\,s^{-2}}$
c) $3.5\ \mathrm{m\,s^{-2}}$
d) $1.2\ \mathrm{m\,s^{-2}}$

48. Taking g as $10\ \mathrm{m\,s^{-2}}$, the work done in lifting 8 blocks, each of weight $12\ \mathrm{kg}$, from the floor onto a shelf $2.8\ \mathrm{m}$ high is

a) $2688\ \mathrm{J}$
b) $268.8\ \mathrm{J}$
c) $26.88\ \mathrm{J}$
d) $2.688\ \mathrm{J}$

49. A car accelerates uniformly from a velocity of $10\ \mathrm{m\,s^{-1}}$ to $60\ \mathrm{m\,s^{-1}}$ in 6 seconds on a straight level road. The distance covered by the car is

a) $150\ \mathrm{m}$
b) $200\ \mathrm{m}$
c) $210\ \mathrm{m}$
d) $215\ \mathrm{m}$

50. Given that the events A and B are such that $P(A'|B) = \dfrac{11}{21}$ and $P(B) = \dfrac{3}{5}$, then $P(A \cap B) =$

a) $\dfrac{5}{7}$
b) $\dfrac{4}{7}$
c) $\dfrac{3}{7}$
d) $\dfrac{2}{7}$

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